Charles' law: how to isolate each quantity

I want V_2

1. Write the law. $$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$
2. The **T**s are **dividing**: each one goes to the other side **multiplying** (cross-multiply). $$V_1 \cdot T_2 = V_2 \cdot T_1$$
3. T_1 is **multiplying** V_2: divide **both sides** by T_1. $$\frac{V_1 \cdot T_2}{T_1} = \frac{V_2 \cdot \cancel{T_1}}{\cancel{T_1}}$$
4. Result: $$\boxed{V_2 = \frac{V_1 \cdot T_2}{T_1}}$$

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4. Result: $$\boxed{T_2 = \frac{V_2 \cdot T_1}{V_1}}$$

I want V_1

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I want T_1

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Shortcut: what is **dividing** on one side goes to the other side **multiplying**, and what is **multiplying** goes to the other side **dividing**.